Agentic Search Efficiency for Scientific Data Discovery: A Three-Path Comparison

Shazzadul Islam, Jaime Cernuda, Xian-He Sun, Anthony Kougkas
Illinois Institute of Technology, Chicago, IL, USA
sislam6@hawk.illinoistech.edu, {jcernudagarcia, sun, akougkas}@illinoistech.edu

***Abstract*—Scientific data search demands reasoning that does not scale: inspecting what each dataset contains, converting between unit conventions, refining queries when initial results fall short. AI agents can automate this, but the cost of running them is rarely examined. A single query can consume nearly a million tokens before producing an answer.**

We present CLIO Search, which implements an observe-decide-act-evaluate cycle for scientific data. Science-aware operators execute deterministic SI conversion and formula normalization alongside BM25 and dense search. CLIO inspects what metadata each corpus provides and adapts accordingly. Federated search spans heterogeneous HPC backends with agentic refinement.

We compare three agentic paths over 10 discovery queries against the National Data Platform (NDP, 341 datasets from NOAA, NASA, ECMWF, and others) using `claude-sonnet-4-20250514`. Path 1 uses a Claude native agent with web search and sub-agent spawning. Path 2 uses NDP-MCP, exposing catalog search via Model Context Protocol. Path 3 uses an agent backed by CLIO Search. All three answer 10/10 queries correctly but cost differs dramatically: the native agent consumes 989K tokens in 50.8 minutes; NDP-MCP consumes 989K tokens in 17.4 minutes; CLIO consumes 474K tokens (-52%) in 4.4 minutes (-91%).

The reduction reflects two factors: smaller context ($\sim$2K tokens per query versus 989K) and fewer reasoning steps (9/10 queries resolve in a single invocation). Moving corpus profiling, unit conversion, and ranking out of the LLM's reasoning loop into a deterministic layer preserves accuracy while halving cost. Agentic search benefits from co-designing specialized retrieval with the agent over raw catalog access.

I. Introduction

Scientific data catalogs are growing faster than human scientists can browse them. The National Data Platform (NDP) [**?**] federates hundreds of repositories; IOWarp and other HPC storage platforms expose billions of objects. AI agents are an increasingly popular way to search these catalogs: the agent reads the query, calls tools, reads results, and produces a ranked list of candidate datasets. This workflow is attractive because it hides the complexity of heterogeneous catalogs behind a single natural-language interface.

However, the cost of running these agents is rarely measured. Each tool call injects results into the LLM's context window; each sub-agent spawn doubles or triples token consumption. When the catalog is large or the query requires iteration, a single query can consume hundreds of thousands of tokens and minutes of wall time. Scaling agentic data discovery to facility-wide catalogs requires understanding where this cost comes from and how to reduce it.

We present CLIO Search [**?**], which implements an observe-decide-act-evaluate cycle for scientific data. Science-aware operators execute deterministic SI conversion and formula normalization alongside BM25 and dense search. CLIO inspects what metadata each corpus provides and adapts accordingly. Federated search spans heterogeneous HPC backends with agentic refinement. In this work, we evaluate one dimension of CLIO's design: its effect on agentic search efficiency over the NDP catalog.

II. Method

Fixed LLM. All three paths use `claude-sonnet-4-20250514` via the Claude Agent SDK, with the same system prompt and the same 10 scientific discovery queries covering temperature, pressure, wind, radiation, and other physical measurements.

Path 1: Claude Native Agent. The agent has access to web search, file read/write, and sub-agent spawning, but no domain-specific tools. It must locate NDP datasets by searching the web, reading pages, and synthesizing findings across sub-agents.

Path 2: NDP-MCP. The agent has access to NDP catalog search via a Model Context Protocol (MCP) [**?**] server. Each call returns raw dataset JSON, which the agent parses and re-ranks in context.

Path 3: CLIO-backed agent. The agent queries CLIO Search, which has indexed the 341 NDP datasets with science-aware metadata extraction: SI unit canonicalization, measurement ranges, and corpus profiling. CLIO returns ranked citations rather than raw JSON.

We measure total token consumption (including sub-agent tokens), wall time, number of tool calls per query, and correctness (whether the agent returns the expected dataset).

III. Results

Table **??** and Figure **??** report the results. All three paths answer 10/10 queries correctly, but their resource consumption differs dramatically. The native agent consumes 989K tokens in 50.8 minutes, averaging 25.6 tool calls per query and spawning 9 sub-agents (563K main + 426K sub-agent

TABLE I
AGENTIC EFFICIENCY ON 10 NDP DISCOVERY QUERIES. Δ IS RELATIVE TO THE CLAUDE NATIVE AGENT.

Method	Tokens	Tok. Δ	Time Δ	Ans.
Claude Native Agent	989,417	—	—	10/10
NDP-MCP	988,954	-0.05%	-65.68%	10/10
CLIO	**474,101**	$\mathbf{-52.08\%}$	$\mathbf{-91.27\%}$	**10/10**

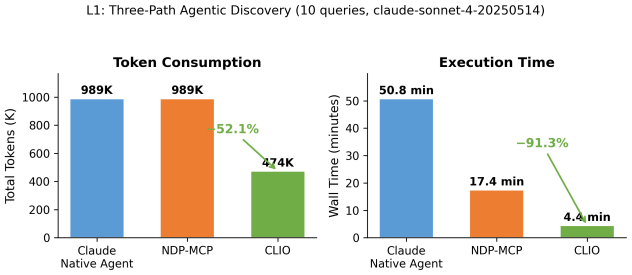

Fig. 1. Token consumption (left) and wall time (right) across 10 queries.

tokens). NDP-MCP consumes 989K tokens in 17.4 minutes with 7.6 calls per query. CLIO consumes 474K tokens (-52%) in 4.4 minutes (-91%) with 2 calls per query, returning $\sim$2K tokens of ranked citations per invocation.

Two factors explain the CLIO reduction. First, *context size*: CLIO returns structured ranked citations rather than raw catalog JSON, avoiding hundreds of kilobytes of documents in the agent's context. Second, *reasoning steps*: 9 of 10 queries resolve in a single invocation. Only one multi-constraint query required four invocations as the agent iteratively refined its search terms.

IV. DISCUSSION

The experiment demonstrates a specific architectural trade-off. Moving corpus profiling, unit conversion, branch selection, and result ranking out of the LLM's reasoning loop and into a deterministic local layer produces substantial savings in both token cost and wall time without sacrificing accuracy. The effect is larger than exposing the same catalog through an MCP tool (NDP-MCP), because the MCP tool still injects raw results that the agent must reason over.

The bottleneck in agentic cost is not the number of tool calls but the *shape* of the data the agent sees. Compact, ranked, citation-style results scale better than raw dumps of catalog JSON. Agentic scientific search benefits from co-designing specialized retrieval with the agent rather than treating the catalog as an opaque tool.

REFERENCES

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